Solution Architecture

Date 05 October 2026

Team ID UNM-293

Project Name FitBuddy-AI fitness plan Generator using Gemini Models

Project ID 2026-2430

Maximum Marks 5 Marks

Solution Architecture Diagram:

High-level representation of the FitBuddy solution architecture, following the Presentation / Client Layer, Application Layer, and Data / Storage Layer structure of the provided template.

Presentation / Client Layer

FitBuddy Web / Mobile User Interface

User enters fitness goal, preferences, available equipment and other required inputs.

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API / Application Gateway

Python API / Backend Server

Receives user requests, validates input and routes requests to the fitness-plan generation service.

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Core Logic Service

FitBuddy Fitness Plan Generator

Processes user requirements and prepares structured prompts for Gemini Models.

Handles fitness-plan generation and formats the AI response for the client.

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Gemini Models / External AI Integration

Gemini API

Generates personalized fitness-plan content from the structured user requirements and application prompts.

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Data / Storage Layer

Application Data Storage

Stores required user inputs, generated plans and project/application data as needed.

Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides the interface through which users enter their fitness goals and preferences and view the generated plan.	Web/Mobile UI, HTML/CSS/JavaScript or project-selected client technology
API / Application Gateway	Receives client requests, validates inputs and connects the user interface with the core FitBuddy service.	Python backend/API framework
Core Logic Service	Processes user requirements, creates prompts, calls Gemini Models and structures the generated fitness plan.	Python, application logic, Gemini API integration
Gemini AI Integration	Provides AI-based fitness-plan generation based on the user's supplied requirements and the application's prompt.	Gemini Models / Gemini API

Component Name	Description / Role in Architecture	Technologies Used
Data / Storage Layer	Stores application data and, where required, user inputs and generated fitness plans for application use.	Project-selected database/storage

Team Member – Assigned Work

Name	Assigned Work
Kaviya sree.T	Responsible for the FitBuddy fitness-plan generation module: collecting and structuring user fitness requirements, preparing prompts for Gemini Models, integrating the Gemini-generated plan into the application, testing the generated output, and documenting this module.